

Faculty of Science, Engineering and Built Environment

SEP291 Engineering Modelling

Deakin University Unit Guide

Trimester 1, 2025

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Welcome

This unit provides a fundamental understanding in the use of mathematical concepts in the design and analysis of engineering systems.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Van Thanh Huynh
Campus: Geelong Waurn Ponds
Email: v.huynh@deakin.edu.au
Phone: +61 3 524 79497
Other members of the team and how to contact them

Details of other staff members involved in this unit will be announced via the unit site.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the Unit have helped them to achieve the learning outcomes:

- Demonstrations of engineering applications for the mathematical methods in solving real world problems.

The following aspects of the Unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- Engineering applications are now included and demonstrated in the first week as well as in the review week at the middle of the trimester.
- Quizzes (Summative Assessments) were implemented. This has helped students follow the learning material. The quizzes have also allowed students to continuously engage and assess their progress throughout the Unit.
- Time for each weekly seminar is now doubled so that students will have more time interacting with the teaching team and discussing weekly topics in a more in-depth manner.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Demonstrate a basic understanding in the fundamental mathematical concepts required to explain the behaviour of engineering systems and familiarity with the analysis tools.	GLO1: Discipline-specific knowledge and capabilities
ULO2	Demonstrate the ability to apply mathematical tools in engineering related systems.	GLO1: Discipline-specific knowledge and capabilities
ULO3	Communicate mathematical concepts to explain the behaviour of engineering systems.	GLO2: Communication GLO4: Critical Thinking

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 40% in the End-of-Unit Assessment.

Brief summary of the hurdle requirement/s	Rationale
To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 40% in the End-of-Unit Assessment.	This unit requires a pass in the End of unit assessment to ensure that each student can demonstrate their learning independently.

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	Problem solving exercise 1
Brief description of assessment task	Demonstrate the understanding of fundamental concepts (Week 1-6)
Detail of student output	Complete a problem solving exercise individually and submit the assignment online. Assignment will consist of questions relating to first 6 weeks of the unit.
Grading and weighting (% total mark for unit)	25% graded

This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: Demonstrate a basic understanding in the fundamental mathematical concepts required to explain the behaviour of Engineering systems and familiarity with the analysis tools. ULO2: Demonstrate the ability to apply mathematical tools in engineering related systems. ULO3: Communicate mathematical concepts to explain the behaviour of Engineering systems.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: This assessment task develops students' knowledge and capabilities in mathematics. GLO2: through using oral, written and interpersonal communication to inform, motivate and effect change. GLO4: through evaluating information using critical and analytical thinking and judgment.
How and when you will receive feedback on your work	Graded assignments will be returned to students via the unit site 2 weeks' after Assignment due date.
When and how to submit your work	Saturday of Week 6 (12 April 2025) 8pm AEST. Form of submission: online in PDF format via the unit site.

Summative assessment task 2

Title	**Problem solving exercise 2**
Brief description of assessment task	Demonstrate an understanding of fundamental concepts (Week 7 -10)
Detail of student output	Complete a problem solving exercise individually and submit the assignment online. Assignment will consist of questions relating to weeks 7 to 10 of the unit.
Grading and weighting (% total mark for unit)	25% - graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: Demonstrate a basic understanding in the fundamental mathematical concepts required to explain the behaviour of Engineering systems and familiarity with the analysis tools. ULO2: Demonstrate the ability to apply mathematical tools in engineering related systems. ULO3: Communicate mathematical concepts to explain the behaviour of Engineering systems.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: This assessment task develops students' knowledge and capabilities in mathematics. GLO2: through using oral, written and interpersonal communication to inform, motivate and effect change. GLO4: through evaluating information using critical and analytical thinking and judgment.
How and when you will receive feedback on your work	Graded assignments will be returned to students via the unit site 2 weeks' after Assignment due date.
When and how to submit your work	Saturday of Week 10 (17 May 2025) 8pm AEST. Form of submission: online submission in PDF format via the unit site.

Summative assessment task 3

Title	**Quizzes (x 5)**
Brief description of assessment task	The quizzes will have questions closely related to seminars delivered in the preceding fortnight. This is an individual assessment task.

Detail of student output	Students are required to complete 5 online problem solving exercises individually. The quizzes will consist of questions relating to weeks 1 to 10 of the unit, and will usually comprise of short answer, long answer and multiple-choice questions. Students will have **one hour complete each quiz.** Only one attempt for each quiz is allowed.
Grading and weighting (% total mark for unit)	10% - (5 quizzes x 2% per quiz) (marked)
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: Demonstrate a basic understanding in the fundamental mathematical concepts required to explain the behaviour of Engineering systems and familiarity with the analysis tools.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: This assessment task develops students' knowledge and capabilities in mathematics
How and when you will receive feedback on your work	The results for each quiz are subject to unit chair's moderation prior to release. Marked quizzes will be returned to students via the unit site within one week of each quiz due date.
When and how to submit your work	The following details the release and due dates for the quizzes: • **Quiz 1:** Release date Week 1: Tuesday 4 March, due on Monday 17 March 2025, Week 3: at 8pm AEDT • **Quiz 2:** Release date Week 3: Tuesday 18 March, due on Monday 31 March 2025 Week 5: 8pm AEDT • **Quiz 3:** Release date Week 5: Tuesday 1 April 2025, due on Monday 14 April 2025, Week 7 at 8pm AEST • **Quiz 4:** Release date Week 7: Tuesday 15 April 2025, due on Monday 5 May 2025, Week 9 at 8pm AEST • **Quiz 5:** Release date Week 9: Tuesday 6 May 2025, due on Tuesday 20 May 2025, Week 11 at 8pm AEST Form of submission: the online quizzes can be accessed and completed via the unit site.

Summative assessment task End-of-unit assessment or Examination

Title	**End-of-Unit Assessment - Written**
Brief description of assessment task	This is an individual assessment task. The task will assess and validate student knowledge of and ability to apply critical thinking techniques to identify and analyse problems from technical and non-technical perspectives. Questions will be based on the topics covered during the entire trimester of study.
How you demonstrate your achievement	2-hour written assessment. A written assessment is a form of assessment that asks you to use text to communicate your knowledge, skill or understanding of a topic. It is completed online. At a set release time, you will be given the assessment instructions and will complete the assessment by a set deadline.
Grading and weighting (% total mark for unit)	40%
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: Demonstrate a basic understanding in the fundamental mathematical concepts. ULO2: Demonstrate the ability to apply mathematical tools in engineering related systems.

This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Students need discipline-specific knowledge and capabilities to answer the exam questions.
How and when you will receive feedback on your work	Students will practise examples of the assessment questions throughout the trimester to help them prepare. Feedback will be provided to the lecture via worked solutions. In addition students will be encouraged to seek feedback during classes as required. Deakin University will release the final assessment results at the stipulated timeframe. Students will receive a mark, which is an indicator of their overall performance in this unit of study.
When and how to submit your work	Students will be required to undertake a timed online assessment during the End-of-unit Assessment Period. You will start and complete at any time within a scheduled 24-hour window, with a 2-hour time limit. Further details will be provided on the unit site. It is the responsibility of students to review their End-of-Unit Assessment timetable when it is released via StudentConnect. Further information including timetable release dates is available at end-of-unit assessments and examinations.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 2 hour seminar per week.
Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including a 1 x 2 hour seminar per week.
Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.
This will include educator guided online learning activities within the unit site.
Teaching, learning and assessment activities for the unit include watching the recorded videos, attending scheduled learning activities (online or on-campus), discussion boards, designated activities, assessment tasks, readings and study time.

The unit involves a lot of mathematical theories. It is strongly encouraged that students engage in the unit from the very early stage. Previous experience with similar units has shown that early engagement with the unit will help build a strong background for the students and facilitate the successful follow-on with the unit.

In addition to spending time studying the seminar questions, students should also undertake hands-on practice with answering such questions. Continuously solving the weekly seminar questions, and from the very early stage of the unit, will help you to complete the unit a lot easier.

Students are also encouraged to actively participate in discussions via the 'Discussions' section in the unit site, watch the recorded video material, read lecture notes, and complete their assignments as published on the unit site.
Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.

Prescribed text(s): Glyn James and Phil Dyke, 2020, Modern Engineering Mathematics, 6th Ed, Pearson Education, Harlow, United Kingdom.

The texts and reading list for SEP291 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

This unit has a prescribed text which is listed above.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

There are a wide variety of textbooks and internet resources useful for this unit, and students are encouraged to utilise all these resources. The following text is recommended, however, there are a number of topics and details in the text that are not relevant to this unit:

TITLE: Advanced Engineering Mathematics
AUTHOR: Erwin Kreyszig
EDITION: 10th Edition
ISBN-10: 0470458364 | ISBN-13: 978-0470458365

Students can access this text via the Deakin Library.

We endeavour to provide you with all the necessary learning materials via the unit site. The underlying concepts are quite common in a number of other texts and also in various internet sites.

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 1 teaching period begins	Monday 3 March 2025
Census date	Monday 31 March 2025
Easter/intra-trimester break	Friday 18 April - Sunday 27 April 2025
Trimester 1 teaching period ends	Friday 23 May 2025
Study period (end-of-unit assessment /examination preparation period)	Monday 26 May - Friday 30 May 2025
End-of-unit assessment and examination period	Monday 2 June - Friday 13 June 2025
Inter-trimester break (the period between trimesters)	Monday 16 June - Friday 4 July 2025
Unit results released	Thursday 3 July 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Assessment due date
1	3 March 2025	Introduction to Ordinary Differential Equations	**Assessment 3: Quiz 1** released 4 March
2#	10 March 2025	Classification and Basic Approaches to solving Differential Equations	
3	17 March 2025	First Order Ordinary Differential Equations	**Assessment 3: Quiz 1** due 17 March **Assessment 3: Quiz 2** released 18 March
4	24 March 2025	Numerical Solution of First Order Ordinary Differential Equations	
5	31March 2025	Linear Constant Coefficient Second Order Ordinary Differential Equations	**Assessment 3: Quiz 2** due 31 March **Assessment 3: Quiz 3** released 1 April
6	7 April 2025	Revision 1 (first-half-trimester topics)	**Assessment task 1: Problem solving exercise** due 12 April
7*~	14 April 2025	Laplace Transform and its Applications (part 1)	**Assessment 3: Quiz 3** due 14 April **Assessment 3: Quiz 4** released 15 April
8	28 April 2025	Laplace Transform and its Applications (part 2)	
9	5 May 2025	Fourier Series and Its Applications	**Assessment 3: Quiz 4** due 5 May **Assessment 3: Quiz 5** released 6 May
10	12 May 2025	Revision 2 (second-half-trimester topics)	**Assessment task 2: Problem solving exercise** due 17 May
11	19 May 2025	All revision	**Assessment 3: Quiz 5** due 20 May
12	26 May 2025		Study period
13-14^	2 June 2025		**End-of-unit assessment**

Labour Day public holiday: **Monday 10 March 2025** - University closed
* Easter/Intra-trimester break: **Friday 18 April - Sunday 27 April 2025** (between weeks 7 and 8)
~ ANZAC Day public holiday: **Friday 25 April 2025** - University closed
^ King's Birthday public holiday: **Monday 9 June 2025** - University closed